

The 2-Adic Sturmian Carry Constant in the Collatz Carry Equation

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Abstract

We study the Sturmian/zero-entropy tower associated to the accelerated odd-only Collatz carry equation. For a valuation word D of length L and valuation sum S , *wall contact* means $2^S - 3^L \mid C_L(D)$, where $C_L(D)$ is the carry sum; wall contacts are exactly the valuation words of integer cycles of the accelerated odd map. Earlier fixed-depth results reduce every fixed nonsingular finite-block family to a finite exact computation, and remain effective for near-critical mechanical blocks; the infinite near-critical tower of continued-fraction levels remains a genuine frontier. This note identifies a native 2-adic object of that tower: the 2-adic Sturmian carry constant $\Xi_\alpha = \sum_{j \geq 0} 3^{-(j+1)} 2^{s_j(c_\alpha)} \in \mathbb{Z}_2$, where c_α is the characteristic Sturmian valuation word of $\alpha = \log_2 3$. For the lower Christoffel blocks X_n of the convergent shells (q_n, p_n) of α , the rotated-lever residues $x_n = -C_n(3^{q_n} - 2^{p_n})^{-1}$ converge to $-\Xi_\alpha$ in $\mathbb{Z}_2$, with the exact valuation law $v_2(x_n + \Xi_\alpha) = p_n - 1$ for upper convergents and $v_2(x_n + \Xi_\alpha) = p_n + p_{n+1} - 1$ for lower convergents. The proof reduces to a floor-comparison lemma for the rational versus irrational mechanical sequences and an ultrametric dominance argument. As a consequence, standard-word singularity — the question of whether a Christoffel block is itself an integer-cycle word — is reduced to a shrinking-window condition on the digits of the single constant Ξ_α , verified empty by exact computation through the shell (190537, 301994). The note does not prove the Collatz conjecture and does not solve the tower; it isolates a precise arithmetic target inside the zero-entropy sector.

1 Introduction

The finite-block obstruction program for the accelerated odd-only Collatz carry equation has produced exact k -block identities, rotated 2-adic levers, and a global fixed-depth finite-reduction theorem: every fixed finite-block family with nonsingular blocks and slopes bounded away from the critical line has all its wall contacts in an explicit finite box [8]. The companion near-critical note [9] showed that this effectiveness survives down any fixed convergent shell of $\alpha = \log_2 3$ — the budget cost of near-criticality is logarithmic, and the lever *gains* strength — but that the infinite tower of continued-fraction levels regenerates the critical problem indefinitely: each rung is effective, and the ladder of rungs is the problem.

This note begins the study of the tower itself, and presents its first theorem-grade structural result. We introduce the 2-adic Sturmian carry constant Ξ_α , the carry series of the infinite characteristic Sturmian valuation word, and prove that the rotated-lever residues of all standard

(lower Christoffel) blocks are truncations of $-\Xi_\alpha$, to an exactly computable depth: the residues converge 2-adically to $-\Xi_\alpha$, with valuation exactly $p_n - 1$ at upper convergents and exactly $p_n + p_{n+1} - 1$ at lower convergents. The proof is elementary and self-contained: a floor-comparison lemma locating the first disagreement between the rational and irrational mechanical sequences, and an ultrametric argument showing the first disagreement dominates.

The consequence for the program is a change of coordinates: the *standard-word singularity* question — is any Christoffel block beyond the first three shells itself an integer-cycle word? — becomes a digit-window condition on one universal 2-adic constant (Section 8), verified empty by exact computation through the shell (190537, 301994). In a sentence:

The tower is not solved, but its residue dynamics are governed by a single 2-adic constant.

Nothing in this note proves, or approaches, the Collatz conjecture, and neither grouped tower products nor unbalanced words nor positive-entropy valuation words are handled here; see Section 11.

2 Collatz carry notation

A *valuation word* is $D = (d_0, \dots, d_{L-1})$, $d_i \geq 1$, with partial sums $s_j = \sum_{i < j} d_i$ (so $s_0 = 0$), $S = s_L$, carry sum

$$C_L(D) = \sum_{j=0}^{L-1} 3^{L-1-j} 2^{s_j}, \quad \text{wall} \quad N_{L,S} = 2^S - 3^L.$$

Wall contact means $N_{L,S} \mid C_L(D)$; wall contacts are exactly the valuation words of cycles of the accelerated odd map $x \mapsto (3x+1)/2^{v_2(3x+1)}$. The critical slope is $\alpha = \log_2 3$, the return path $R_j = s_j - j\alpha$. For a finite block X : $q = |X|$, $S_X = \sum X$, $\delta_X = 3^q - 2^{S_X}$ (odd), and the rational block value

$$Q(X) = -\frac{C(X)}{\delta_X},$$

which is the value of the (rational) cycle with valuation word X ; X is *singular* when $\delta_X \mid C(X)$, i.e. when $Q(X) \in \mathbb{Z}$ and X is an integer-cycle word. For the standard blocks below we write $C_n = C(X_n)$, $\delta_n = 3^{q_n} - 2^{p_n}$, $Q_n = -C_n/\delta_n$.

3 Sturmian/Christoffel conventions

Set $\theta = \alpha - 1 = \log_2(3/2) \in (0, 1)$. The convergent shells of α are (q_n, p_n) with p_n/q_n the continued-fraction convergents of α :

$$(1, 1), (1, 2), (2, 3), (5, 8), (12, 19), (41, 65), (53, 84), (306, 485), (665, 1054), (15601, 24727), \dots$$

and $p'_n := p_n - q_n$ makes p'_n/q_n the convergents of θ , with $\gcd(p'_n, q_n) = 1$. The slope defect is $R_n = p_n - q_n\alpha = p'_n - q_n\theta$; we call the convergent *upper* when $R_n > 0$ (i.e. $p_n/q_n > \alpha$) and *lower* when $R_n < 0$; signs alternate. Classical best-approximation facts used below:

$$\frac{1}{q_n + q_{n+1}} < |R_n| < \frac{1}{q_{n+1}}, \quad q_{n+1}|R_n| + q_n|R_{n+1}| = 1, \quad (1)$$

the latter from $|q_{n+1}p_n - q_n p_{n+1}| = 1$ with the α -terms cancelling.

The *lower mechanical (Christoffel) block* of shell (q_n, p_n) , over valuation letters $\{1, 2\}$, is

$$X_n = (d_0, \dots, d_{q_n-1}), \quad d_i = 1 + \left\lfloor \frac{(i+1)p'_n}{q_n} \right\rfloor - \left\lfloor \frac{ip'_n}{q_n} \right\rfloor,$$

so that $|X_n| = q_n$ and $S(X_n) = p_n$. The *characteristic Sturmian valuation word* is $c_\alpha = (d_0, d_1, \dots)$ with $d_j = 1 + \lfloor (j+1)\theta \rfloor - \lfloor j\theta \rfloor$. For any word w we write $s_j(w) = \sum_{i < j} d_i(w)$. The defining formulas telescope:

$$s_j(c_\alpha) = j + \lfloor j\theta \rfloor \quad (j \geq 0), \quad s_j(X_n) = j + \lfloor j\theta_n \rfloor \quad (0 \leq j \leq q_n), \quad \theta_n := \frac{p'_n}{q_n}. \quad (2)$$

4 The periodic-carry identity

Definition 4.1. For an infinite valuation word w (all letters ≥ 1 , so $s_j(w) \geq j \rightarrow \infty$), define

$$\Xi(w) = \sum_{j \geq 0} 3^{-(j+1)} 2^{s_j(w)} \in \mathbb{Z}_2.$$

The series converges in $\mathbb{Z}_2$: $3^{-(j+1)}$ is a 2-adic unit and v_2 of the j -th term is $s_j(w) \rightarrow \infty$. The *2-adic Sturmian carry constant* is $\Xi_\alpha := \Xi(c_\alpha)$.

Theorem 4.2 (periodic-carry identity). *Let X be a finite block with $q = |X|$, $S_X = \sum X$, and let X^∞ be its infinite periodic extension. Then*

$$\Xi(X^\infty) = \frac{C(X)}{3^q - 2^{S_X}} = \frac{C(X)}{\delta_X} = -Q(X).$$

In particular, for the standard blocks,

$$x_n := -C_n \delta_n^{-1} = Q_n = -\Xi(X_n^\infty) \in \mathbb{Z}_2,$$

the rational value Q_n embedded in $\mathbb{Z}_2$.

Proof. Write $j = tq + r$, $0 \leq r < q$; periodicity gives $s_j(X^\infty) = tS_X + s_r(X)$. Grouping by periods,

$$\Xi(X^\infty) = \sum_{t \geq 0} \sum_{r=0}^{q-1} 3^{-(tq+r+1)} 2^{tS_X + s_r} = \left(\sum_{r=0}^{q-1} 3^{-(r+1)} 2^{s_r} \right) \sum_{t \geq 0} (3^{-q} 2^{S_X})^t.$$

The geometric series converges in $\mathbb{Z}_2$ because $v_2(3^{-q} 2^{S_X}) = S_X \geq 1$, with sum $(1 - 3^{-q} 2^{S_X})^{-1}$; and $C(X) = \sum_r 3^{q-1-r} 2^{s_r} = 3^q \sum_r 3^{-(r+1)} 2^{s_r}$. Multiplying, $\Xi(X^\infty) = 3^{-q} C(X) (1 - 3^{-q} 2^{S_X})^{-1} = C(X) / (3^q - 2^{S_X}) = C(X) / \delta_X = -Q(X)$. The sign bookkeeping for the standard blocks follows from $\delta_n = 3^{q_n} - 2^{p_n}$ and $Q_n = -C_n / \delta_n$. $\square$

5 The floor-comparison lemma

The periodic extension also obeys the floor formula *globally*: for $j = tq_n + r$,

$$s_j(X_n^\infty) = tp_n + r + \lfloor r\theta_n \rfloor = j + \lfloor tp'_n + r\theta_n \rfloor = j + \lfloor j\theta_n \rfloor.$$

By (2), comparing X_n^∞ with c_α is exactly comparing the floor sequences of the rational rotation θ_n and the irrational rotation θ : *the first partial-sum disagreement is the first $j \geq 1$ with $\lfloor j\theta_n \rfloor \neq \lfloor j\theta \rfloor$.*

Lemma 5.1 (floor-comparison / agreement-depth). *Let $j_n^* = \min\{j \geq 1 : \lfloor j\theta_n \rfloor \neq \lfloor j\theta \rfloor\}$. Then:*

1. *If $R_n > 0$ (upper): $j_n^* = q_n$, and the partial-sum depths at the mismatch are*

$$s_{j^*}(X_n^\infty) = p_n, \quad s_{j^*}(c_\alpha) = p_n - 1.$$

2. *If $R_n < 0$ (lower): $j_n^* = q_n + q_{n+1}$, and*

$$s_{j^*}(X_n^\infty) = p_n + p_{n+1} - 1, \quad s_{j^*}(c_\alpha) = p_n + p_{n+1}.$$

In letters, the agreement length is $j_n^ - 1$ (the first differing letter has index $j_n^* - 1$); the valuation law of Section 7 lives at the first partial-sum mismatch j_n^* .*

Proof. Write $\theta = \theta_n + \eta$, so $\eta = -R_n/q_n$, and let $m_j = jp'_n \bmod q_n$, so $\{j\theta_n\} = m_j/q_n$; coprimality gives $m_j \geq 1$ for $1 \leq j < q_n$, and $m_{q_n} = 0$.

(1) *Upper:* $\eta < 0$, $|\eta| = R_n/q_n$. A mismatch at j means $\lfloor j\theta_n + j\eta \rfloor < \lfloor j\theta_n \rfloor$, i.e. $\{j\theta_n\} < j|\eta|$, i.e. $m_j < jR_n$. For $1 \leq j < q_n$ this fails: $m_j \geq 1$ while $jR_n < q_n R_n < q_n/q_{n+1} < 1$ by (1). At $j = q_n$ it holds: $m_{q_n} = 0 < q_n R_n$, and since $-1 < q_n \eta < 0$, $\lfloor q_n \theta \rfloor = p'_n - 1$. Depths: $s_{q_n}(X_n^\infty) = q_n + p'_n = p_n$ and $s_{q_n}(c_\alpha) = q_n + p'_n - 1 = p_n - 1$.

(2) *Lower:* $\eta > 0$, $\eta = |R_n|/q_n$. A mismatch at j means $\{j\theta_n\} + j\eta \geq 1$, i.e. $m_j + j|R_n| \geq q_n$. Since $m_j \leq q_n - 1$, this needs $j|R_n| \geq 1$, hence $j > q_{n+1}$ by (1) (as $q_{n+1}|R_n| = 1 - q_n|R_{n+1}| < 1$). Two sign facts: for a lower convergent the next convergent lies above, so $q_{n+1}p_n - q_n p_{n+1} = -1$, giving $m_{q_{n+1}} = q_{n+1}p'_n \bmod q_n = q_n - 1$; and for $j = q_{n+1} + i$ with $1 \leq i \leq q_n - 1$, $m_j = (ip'_n \bmod q_n) - 1 \leq q_n - 2$ (coprimality gives $ip'_n \not\equiv 0$). Now in the range $q_{n+1} \leq j < q_n + q_{n+1}$ there is no mismatch: at $j = q_{n+1}$, $m_j + j|R_n| = (q_n - 1) + (1 - q_n|R_{n+1}|) < q_n$; and for $q_{n+1} < j < q_n + q_{n+1}$, $j|R_n| < (q_n + q_{n+1})|R_n| = q_n|R_n| + 1 - q_n|R_{n+1}| < 2$ (using $q_n|R_n| < 1$), so $m_j + j|R_n| < (q_n - 2) + 2 = q_n$. At $j^* = q_n + q_{n+1}$ the mismatch occurs: $m_{j^*} = m_{q_{n+1}} = q_n - 1$ and $m_{j^*} + j^*|R_n| = q_n + q_n(|R_n| - |R_{n+1}|) > q_n$. Depths: $j^*\theta_n = p'_n + p'_{n+1} - 1/q_n$ gives $s_{j^*}(X_n^\infty) = j^* + p'_n + p'_{n+1} - 1 = p_n + p_{n+1} - 1$; and $j^*\eta = |R_n| + 1/q_n - |R_{n+1}|$ gives $\lfloor j^*\theta \rfloor = p'_n + p'_{n+1}$, so $s_{j^*}(c_\alpha) = p_n + p_{n+1}$. $\square$

Remark 5.2. The lemma is consistent with the classical fact that the periodic approximant of a one-sided best rational approximation agrees with the characteristic word for $q_n + q_{n+1} - O(1)$ letters [6, 7]; the proof above is self-contained and pins the constants and both partial-sum depths exactly, which the valuation law requires. The asymmetry is structural: c_α is the limit of *lower* mechanical words, so lower-convergent periodic extensions agree with it one full continued-fraction level longer.

6 Ultrametric first-mismatch dominance

Lemma 6.1 (ultrametric dominance). *Let u, v be infinite valuation words whose partial sums first differ at index j^* , with depths $A = \min(s_{j^*}(u), s_{j^*}(v))$ and $A' = \max(\cdot, \cdot) > A$. Then*

$$v_2(\Xi(u) - \Xi(v)) = A.$$

Proof. $3^{-(j+1)}$ is a 2-adic unit, so term j of $\Xi(u) - \Xi(v) = \sum_j 3^{-(j+1)}(2^{s_j(u)} - 2^{s_j(v)})$ has $v_2 = \min(s_j(u), s_j(v))$ when the sums differ, and vanishes when they agree. The first nonzero term, $j = j^*$, has $v_2 = A$ exactly. For $j > j^*$, since letters are ≥ 1 , $s_j(u) \geq s_{j^*}(u) + (j - j^*)$ and

likewise for v , so every later nonzero term has $v_2 \geq A + (j - j^*) \geq A + 1$. The tail therefore has $v_2 \geq A + 1$, and by the ultrametric inequality the sum has $v_2 = A$: later terms cannot cancel the leading one. (In our application the depths are consecutive, $\{A, A + 1\}$, and the leading term is $3^{-(j^*+1)}(\pm 2^A \mp 2^{A+1}) = \mp 3^{-(j^*+1)} 2^A$.) $\square$

7 The main convergence theorem

Theorem 7.1 (Sturmian carry convergence law). *Let X_n be the lower Christoffel block of the convergent shell (q_n, p_n) of $\alpha = \log_2 3$, and $x_n = -C_n \delta_n^{-1} \in \mathbb{Z}_2$. Then*

$$x_n \longrightarrow -\Xi_\alpha \text{ in } \mathbb{Z}_2,$$

and, exactly,

$$v_2(x_n + \Xi_\alpha) = \begin{cases} p_n - 1, & R_n > 0 \text{ (upper convergents)}, \\ p_n + p_{n+1} - 1, & R_n < 0 \text{ (lower convergents)}. \end{cases}$$

Proof. By Theorem 4.2, $x_n + \Xi_\alpha = -(\Xi(X_n^\infty) - \Xi(c_\alpha))$. Lemma 5.1 locates the first partial-sum mismatch of X_n^∞ against c_α and computes both depths; Lemma 6.1 says the valuation is the smaller depth: $\min(p_n, p_n - 1) = p_n - 1$ in the upper case and $\min(p_n + p_{n+1} - 1, p_n + p_{n+1}) = p_n + p_{n+1} - 1$ in the lower case. Convergence follows since the valuations tend to infinity. $\square$

Remark 7.2. The proof does not use the alternating standard-word recursion $X_{n+1} = X_n^{a_{n+1}} X_{n-1}$ (lower R_n) / $X_{n-1} X_n^{a_{n+1}}$ (upper R_n): it depends only on the mechanical floor formula (2) and Lemma 5.1, which removes the recursion's order convention from the trust chain. The carry cocycle $M(W) = \begin{pmatrix} 2^{S_W} & C(W) \\ 0 & 3^{q_W} \end{pmatrix}$, $M(UV) = M(U)M(V)$, together with that recursion, remains the computational engine: it evaluates C_n and $x_n \bmod 2^M$ in $O(\log)$ matrix multiplications and was verified against direct construction through the shell (665, 1054).

8 The singularity window criterion

Singularity of the standard block, $\delta_n \mid C_n$, is equivalent to $Q_n \in \mathbb{Z}$. Since $Q_n = x_n$ (Theorem 4.2) and $x_n \equiv -\Xi_\alpha \pmod{2^{\ell_n}}$ with

$$\ell_n = \begin{cases} p_n - 1, & R_n > 0, \\ p_n + p_{n+1} - 1, & R_n < 0 \end{cases}$$

(Theorem 7.1), an integer Q_n is pinned by its residue and bounded by its archimedean size. For mechanical blocks the internal heights satisfy $r_j \in (-1, 0]$ (zero internal rise; [9]), so $|Q_n| = \frac{1}{3} \sum_j 2^{r_j} / |2^{R_n} - 1| \leq \frac{2q_n}{3 \ln 2} |R_n|^{-1}$, and with (1),

$$|Q_n| \leq B_n := q_n (q_n + q_{n+1}), \quad \log_2 B_n = O(\log q_{n+1}) = o(p_n). \quad (3)$$

Proposition 8.1 (window criterion). *Let w_n be the centered representative of $-\Xi_\alpha \bmod 2^{\ell_n}$, i.e. the unique integer $w_n \equiv -\Xi_\alpha \pmod{2^{\ell_n}}$ with $|w_n| < 2^{\ell_n-1}$. If X_n is singular and $2B_n + 1 < 2^{\ell_n}$, then*

$$Q_n = w_n \quad \text{and in particular} \quad |w_n| \leq B_n \quad (\text{the window event}).$$

The window event is necessary, not sufficient: it must be completed by the exact integrality check $C_n = -w_n \delta_n$.

Proof. If $Q_n \in \mathbb{Z}$ then $Q_n \equiv -\Xi_\alpha \pmod{2^{\ell_n}}$ and $|Q_n| \leq B_n < 2^{\ell_n-1}$, so Q_n is the centered representative. $\square$

Remark 8.2 (level-4 caveat). At the first nonsingular shell (5, 8) the criterion is degenerate: $\ell_4 = 7$ and $2B_4 + 1 > 2^{\ell_4-1}$, the window is not small relative to the modulus, and indeed the computed centered residue lies *inside* the window while $Q_4 = 319/13 \notin \mathbb{Z}$. The criterion bites from the next level onward, where the computed relative window sizes $(2B_n + 1) 2^{-\ell_n}$ collapse:

$$2^{-74} (12, 19), \quad 2^{-52} (41, 65), \quad 2^{-554} (53, 84), \quad 2^{-466} (306, 485), \quad 2^{-25757} (665, 1054), \dots$$

and the computed centered residues lie outside the window at every level checked.

Conjecture 8.3 (standard-word nonsingularity). $\delta_n \nmid C_n$ for all $n \geq 4$. *Equivalently, in the digit-window formulation: for every $n \geq 5$ the truncations of $-\Xi_\alpha$ never complete a window event together with the exact integrality condition, and level 4 is checked directly.*

9 Known-cycle calibration and the -17 shell

The first three shells are singular, and are exactly the known small cycles:

$$(1, 1) : Q = -1, \quad (1, 2) : Q = +1, \quad (2, 3) : Q = -5,$$

with the -7 cycle appearing as the rotation class of level 3: the conjugate (2, 1) of the Christoffel word (1, 2) has $C = 7$, $\delta = 1$, value -7 . Rotations of a singular Christoffel word are singular with the conjugate cycle values, so the exception list of Conjecture 8.3 is closed under rotation, and $n \geq 4$ is the correct and sharp starting point.

The semiconvergent shell $(7, 11) = (2, 3) + (5, 8)$ has wall $|3^7 - 2^{11}| = 139$. Its mechanical block $(1, 2, 1, 2, 1, 2, 2)$ is *nonsingular* ($C = 3767$, $139 \nmid 3767$), while the -17 cycle word

$$(1, 1, 1, 2, 1, 1, 4)$$

lies on the same shell, is *unbalanced* (it contains the letter 4), and is a wall contact. Hence

shell singularity is not the same as mechanical-word singularity,

and Conjecture 8.3 speaks only of the mechanical words: the unbalanced residents of a shell are a separate, demonstrably nonempty, arithmetic population.

10 Computational ledger

All computations are exact (arbitrary-precision integer arithmetic): direct construction of X_n and termwise carry sums for $q_n \lesssim 10^3$; the transfer-matrix cocycle with the alternating recursion for deep shells, verified against direct construction through (665, 1054); $\Xi_\alpha \bmod 2^M$ from a characteristic-word prefix evaluated with a certified rational sandwich for θ and unambiguous-floor assertions ($M = 4096$ in the final run). The table reports the ledger; “ v_2 ” records the verified valuation of $x_n + \Xi_\alpha$ against Theorem 7.1.

n	shell (q_n, p_n)	sign of R_n	ℓ_n	singular?	$\log_2 Q_n $	status
1	(1, 1)	—	—	yes	$Q = -1$	the -1 cycle
2	(1, 2)	+	—	yes	$Q = +1$	the $+1$ cycle
3	(2, 3)	—	—	yes	$Q = -5$	the -5 cycle (-7 : rotation)
4	(5, 8)	+	7	no	4.6	$v_2 = 7$ verified exactly
5	(12, 19)	—	83	no	7.8	$v_2 = 83$ verified exactly
6	(41, 65)	+	64	no	≈ 10	$v_2 = 64$ verified exactly
7	(53, 84)	—	568	no	≈ 12	$v_2 = 568$ verified exactly
8	(306, 485)	+	484	no	≈ 16	$v_2 = 484$ verified exactly
9	(665, 1054)	—	25780	no	≈ 22	consistent to 4096 bits
10	(15601, 24727)	+	24726	no	≈ 27	predicted
11	(31867, 50508)	—	176250	no	≈ 30	predicted
12	(79335, 125743)	+	125742	no	≈ 33	predicted
13	(111202, 176251)	—	478244	no	≈ 33	predicted
14	(190537, 301994)	+	301993	no	≈ 39	predicted

Singularity was tested directly (exact divisibility) at every level shown: levels 1–3 singular, levels 4–14 nonsingular, i.e. through the shell (190537, 301994) with $|\delta| \sim 2^{301970}$. The internal quantities of Lemma 5.1 (j_n^* , both partial-sum depths, and the sign fact $m_{q_{n+1}} = q_n - 1$) were verified independently at levels 4–9, including $j^* = 16266 = 665 + 15601$ with depths (25780, 25781) at level 9. These computations are evidence for Conjecture 8.3 at the checked levels only; Theorem 7.1 itself is proved for all n .

11 Limitations and next targets

This note does not prove the Collatz conjecture. It does not solve the Sturmian tower: it proves the convergence law for pure standard blocks and reduces their singularity to a digit-window condition, but it does not handle grouped tower products $X_n^a X_m^b$ (whose residues mix two levels), unbalanced low-complexity residents (where the -17 cycle lives), or positive-entropy valuation words, which lie outside the mechanical framework entirely.

Next targets, in order: (1) attack Conjecture 8.3 through the window criterion, which now requires only an effective non-integrality statement for the truncations of Ξ_α ; (2) study the digit behavior of Ξ_α itself — normality, recurrence of windows, and the structure inherited from the Sturmian recursion; (3) extend the grouped-sector ledgers with boxes priced by the global fixed-depth theorem; (4) analyze unbalanced semiconvergent residents such as the -17 shell, where word-level and shell-level arithmetic demonstrably diverge; (5) connect zero-entropy results to broader cycle-shape constraints, without overclaiming: each of these is a step inside the tower, and the tower is the frontier, not the finish.

Tier ledger. Tier 1: Theorem 4.2, Lemma 5.1, Lemma 6.1, Theorem 7.1, Proposition 8.1, and the calibration facts of Section 9. Tier 3: the computational ledger of Section 10 (evidence within checked levels). Tier 4: Conjecture 8.3 and the broader tower program.

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